Figure 2: Networks with both spike frequency adaptation (SFA) and short-term synaptic depression (STD) stay near the edge of chaos, even as the stimulus changes. We simulated a network with 300 neurons using a random, sparse ($d = 1/3$), connection matrix W (a), with Gaussian-distributed non-zero weights (b) with modest inhibitory dominance ($\tilde{\mu}_I = \frac{4}{3}\tilde{\mu}_E$), with an eigenspectrum (c) of the Jacobian, $J = \frac{1}{\tau_d}(W - I)$, that has many unstable values right of the imaginary axis, necessitating stability-promoting mechanism(s). We used a hard sigmoid nonlinearity ϕ with rounded corners and range $[0, 1]$ to further enforce Dale's law on synaptic output, unlike $\tanh!(x)$. We show a comparison of this network with SFA only (d–m, left) to the same network with both SFA+STD (n–w, right). With only SFA, a random stimulus (d) applied to 15% of excitatory neurons causes an elevated, chaotic, positive local largest Lyapunov exponent (i). In comparison, the combination of SFA+STD still allows fluctuations, but remains centered near the edge of chaos ($\lambda_1^{\mathrm{Lyp}} = 0$; s, dashed line). (e,o) Each neuron's dendrite x_i integrates excitatory, inhibitory, and external input with a dendritic time constant $\tau_d = 100, \mathrm{ms}$. Each neuron has three time constants of SFA ($\tau * k = 1, 2, 3 = [0.1, 1, 10], \mathrm{s}$), which were summed $\sum * k = 1^K a_{ik}$ for plotting (g,q), that subtractively reduce the spike rate r_i, while STD multiplicatively reduces the synaptic output $b_j r_j$ (f,p). Halfway through the stim period (j,t) and halfway through the following no-stim period (k,u), we plot the effective connectivity matrix J_{eff} (Eq. 7). The negative diagonal is due to $-I$. Importantly, the vertical white bands are due to the dynamic interaction of SFA, STD, and nonlinearity ϕ with the static connectivity W. We show the eigenspectra of the full Jacobians (including SFA & STD dynamics) during stim (l,v) and no-stim (m,w), and can see that these eigenvalues are largely left of the imaginary axis. The eigenspectra (l–w) and effective connectivity (j–u) continuously modulate in response to internal dynamics and external inputs. We replicated this stim vs. no-stim comparison of stability across four conditions: no adaptation (x), SFA only (y), STD only (z), and SFA+STD (zz). For each condition, we simulated 25 random, sparsely connected networks (same parameters as a–w), but varied the fraction f of excitatory neurons from 0.4 to 0.6, equivalent to E:I ratios of 2:3 to 3:2. Networks with no adaptation (x) had increased chaos during stimulation ($p = 0.035$, two-tailed Wilcoxon signed-rank test, median difference 0.46). Networks with SFA only (y) also had increased chaos ($p = 0.0012$, median difference 0.38). Networks with STD only (z) did not have a significant change ($p = 0.31$, median difference -0.13), and networks with SFA+STD (zz) did not have a significant change ($p = 0.47$, median difference 0.04). Notably, SFA+STD kept random, sparse networks lacking E–I balance near the edge of chaos, even during random excitatory stimuli.